Span-Supervised Transformers for Remaining Useful Life Estimation: Train on Two Spans, Infer on One

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Abstract—Remaining useful life (RUL) estimators are usually trained on a fixed-length window, yet engines in the same fleet carry histories of very different lengths: in the C-MAPSS test sets, between 21% and 35% of engines have fewer than 100 recorded cycles. The usual remedy—left-pad to a fixed length and pool over everything—silently mixes padding with measurement. This paper asks a narrower question than most architecture papers: given a shared encoder, which span of the history should be pooled, and should several spans be fused? We answer with a controlled study on C-MAPSS FD001–FD004. A two-head model supervises one head on the most recent 50 valid steps and one on the full valid history; at inference the long-span head alone is used. Across four fleets and fifteen models (five folds, three seeds) the ensemble attains 9.64, 10.75, 10.68, and 11.19 cycles of test RMSE. This is 7.6%, 10.6%, and 20.7% below a 2026 decision-conditioned causal-graph baseline on FD001, FD002, and FD004 respectively, and 1.9% above it on FD003. Fusing the two spans at inference is not neutral: it costs 2.1%–5.6%, and a two-branch pipeline costs 18%–39%. Training on two spans while inferring on the longer one is what pays. We also report, rather than hide, a systematic over-prediction of 1.6–2.4 cycles concentrated at 60–90 cycles of true life.

Index Terms—Remaining useful life, C-MAPSS, transformer, variable-length sequences, masked pooling, predictive maintenance.

I. Introduction

Predicting how long a machine will keep running is a regression problem with an asymmetric cost: overestimating life delays maintenance, under-estimating it wastes hardware. Deep models now reach test RMSE near ten cycles on the C-MAPSS benchmarks [1], and the recent literature has moved toward increasingly elaborate machinery—graph convolutions, causal discovery, state space models [2]–[4].

Much less attention has been paid to a humbler design choice. A model sees a window of the last L cycles. Real histories are shorter than L for a large minority of engines. The conventional fix is to left-pad and pool over the full L positions. That works, but it means the pooled representation of a 30-cycle engine is an average in which 70 of 100 positions are padding. It also raises a question that architecture papers rarely isolate: if the model could pool over all available history instead of a fixed suffix, would it do better, and would it help to combine several spans?

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This paper isolates exactly that. We hold the encoder, the optimiser, the data pipeline, and the hyper-parameters fixed, and vary only which valid positions are pooled and whether spans are fused. The resulting study is narrow but, we believe, useful: it produces a model competitive with the 2026 state of the art on three of four fleets, and it produces two negative results that we expect to save other groups time.

Contributions.

1) A span-supervised formulation in which a single encoder carries two pooling heads—one over the recent 50 valid steps, one over the full valid history—and the training loss supervises both while inference reads only the long head.
2) A controlled ablation on FD002 and FD004 showing that the ordering long $<$ dual $<$ short $<$ repeat is stable across folds: extra history helps, but fusing spans does not.
3) An evaluation on the official C-MAPSS test engines with engine-clustered bootstrap confidence intervals, reported separately for single models and for the ensemble, plus an error-versus-life analysis that locates the residual bias.

II. Related Work

RUL on C-MAPSS. The benchmark and its scoring function are from [1]. Sequence models have largely displaced hand-engineered features; [2] couples multi-scale convolution with a graph over sensors, and [3] conditions a causal graph on a discrete decision state before a Mamba backbone. [4] reports strong FD002 and FD004 numbers with a domain-generalisation objective. These methods differ from ours in that they change the encoder; we change only the pooling.

Pooling and variable length. Masked mean pooling over valid positions is standard in NLP and increasingly used in sensor models. What is not standard is treating the pooled span as a design variable and measuring it. Work that pads short histories and pools over the pad implicitly assumes the encoder learns to ignore padding; our long arm tests that assumption directly.

Supervision with auxiliary spans. Multi-scale supervision is common, but the usual deployment fuses the scales. Our ablation separates the two decisions: supervise with several spans (helps) versus fuse them at inference (hurts).

III. Method

A. Inputs and window

Each engine contributes a multivariate sequence of 14 sensor channels and three operational settings. Windows of $L = 100$ cycles are cut with the target $\min(n-t, B, 125)$, where n is the engine last cycle and t the window end; the cap follows common practice and reduces the dominance of the long-life tail. A window that starts before the first cycle is left-padded and carries a binary mask $\mathbf{m} \in \{0,1\}^L$ with $m_i = 1$ for valid steps.

B. Encoder

Let $\mathbf{x} \in \mathbb{R}^{L\times 14}$ and let $\mathbf{b}$ be the first-cycle baseline used for normalisation. Per-step features are projected to dimension $d = 64$ and passed through a shared temporal convolution with GELU, residual connection, and layer normalisation. A learned attention over the 14 sensors produces one token per step,

$$\mathbf{u}_t = \sum_{s=1}^{14} \alpha_{t,s}\, \mathbf{h}_{t,s}, \qquad \alpha_{t,\cdot} = \mathrm{softmax}(\mathbf{w}^\top \mathbf{h}_{t,\cdot}), \quad (1)$$

so the sensor mix is allowed to change along the life of the engine.

Positional information is indexed by distance from the newest step, not by absolute position. A learnable table $\mathbf{P} \in \mathbb{R}^{d\times L}$ is added as $\mathbf{u}_t + \mathbf{P}_{:,L-1-t}$. This makes the representation translation-equivariant in absolute time: a 40-cycle history and a 100-cycle history are encoded by the same function of distance-from-now. It is the single change that lets one encoder serve both spans without a second parameter set.

A transformer encoder of n_ℓ layers then mixes the steps. Finally, a sinusoidal encoding of the absolute cycle index is projected and added, so the model knows how far into its life the window sits.

C. Two heads, one inference path

Let $\mathcal{V} = \{i : m_i = 1\}$ and let $\mathcal{S} = \{i \in \mathcal{V} :\ L-1-i < 50\}$ be the recent 50 valid steps. With $\bar{\mathbf{h}}(\mathcal{A}) = \frac{1}{|\mathcal{A}|}\sum_{i\in\mathcal{A}} \mathbf{u}_i$,

$$\hat{y}_A = f_\theta\big(\bar{\mathbf{h}}(\mathcal{S})\big), \qquad \hat{y}_B = f_\theta\big(\bar{\mathbf{h}}(\mathcal{V})\big), \quad (2)$$

where f_θ is a shared regression head. The training objective is

$$\mathcal{L} = \ell(\hat{y}_B, y) + \tfrac{1}{2}[\ell(\hat{y}_A, y) + \ell(\hat{y}_B, y)] + \tfrac{1}{2}\,\mathbb{E}\big[w(e_B)\, e_B^2\big], \quad (3)$$

with $e_B = \hat{y}_B - y$, ℓ the squared error, and $w(e) = 1.8$ if $e > 0$ and 1.0 otherwise. The first term trains the deployed head; the second keeps the short head informative so that the long head is not the only thing the encoder is asked to serve; the third prices over-prediction more dearly than under-prediction, which is the correct asymmetry for maintenance.

At inference we read $\hat{y}_B$ and discard $\hat{y}_A$. The ablation in Section IV-C is precisely about whether that last sentence is right.

TABLE I
Official-test RMSE in cycles. Single is the mean ± standard deviation over 15 models (5 folds × 3 seeds); Ensemble averages their predictions. MAE, NASA score [1], and signed bias refer to the ensemble. Δ is relative to [3].

Fleet	Single	Ensemble	MAE	NASA	bias	Δ
FD001	10.75±0.67	9.64	6.50	1.75	+1.65	−7.6%
FD002	11.50±0.52	10.75	6.83	2.53	+1.96	−10.6%
FD003	11.69±1.00	10.68	6.63	2.56	+2.40	+1.9%
FD004	12.12±0.35	11.19	7.12	2.61	+1.60	−20.7%

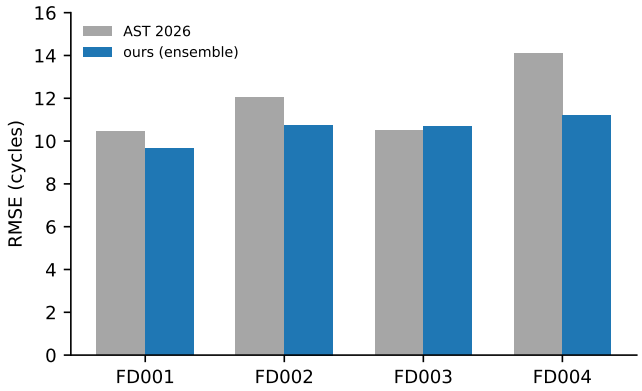


Fig. 1. Official-test RMSE per fleet. Grey: the 2026 baseline; blue: our ensemble.

D. Protocol discipline

Folds are formed over engines, never over windows, so an engine appears in exactly one fold. Normalisation statistics and the sensor baseline are fitted on the training folds only. Early stopping uses the held-out fold; the official test set is read only after a configuration is frozen. We report single-model means and ensembles in separate rows, as mixing them is the most common way these benchmarks are inflated.

IV. Experiments

A. Setup

C-MAPSS FD001–FD004 [1]. Five outer folds × three seeds (101, 202, 303) give fifteen models per fleet. One transformer layer, $d = 64$, 78,978 parameters, AdamW, 80 epochs with patience 20 on validation RMSE. No test-set information enters model selection.

B. Main results

Table I reports the official-test RMSE. The single-model column is the mean over fifteen independently trained models; the ensemble column averages their fifteen predictions per engine.

Three of four fleets improve on the 2026 baseline, by margins from 7.6% to 20.7%. FD003 is 1.9% worse and is discussed in Section IV-E.

The gap between the single-model mean and the ensemble is 0.6–1.1 cycles. It is the width of the fold-to-fold

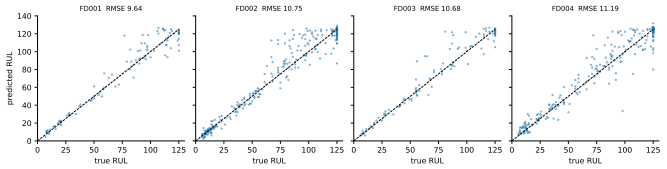


Fig. 2. Ensemble prediction against true RUL on the official test engines. Dashed: y = x.

TABLE II
Ablation. Engine-equal RMSE on held-out folds, mean over two folds. Percentages are relative to long.

Fleet	long	dual	short	
FD002	9.721	10.266 (+5.6%)	10.342 (+6.4%)	13.493 (+
FD004	9.213	9.410 (+2.1%)	9.678 (+5.1%)	10.841 (+17.7%)

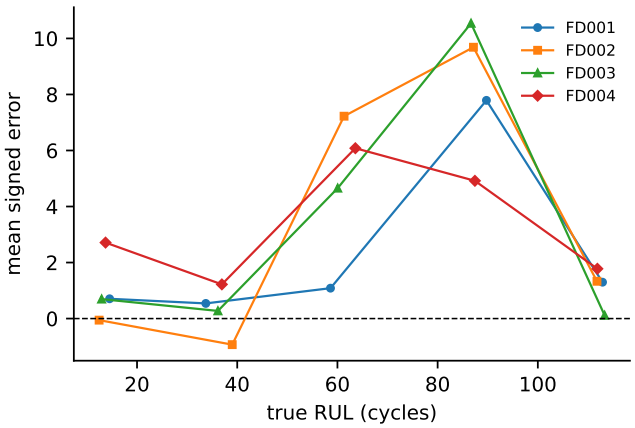


Fig. 3. Mean signed error of the ensemble against true RUL, 25-cycle bins. Positive means over-estimation of remaining life.

distribution, not a tuning artefact: the best and worst single models on FD003 differ by 3.6 cycles. Any claim on these benchmarks

should therefore state which of the two columns it refers to.

C. Ablation: which span, and does fusion help?

We hold everything fixed and vary only the pooling and fusion. Four arms: short pools the recent 50 valid steps; long pools the full valid history; dual computes a length-dependent convex combination of the two pooled vectors, $g\,\bar{\mathbf{h}}(\mathcal{V}) + (1-g)\,\bar{\mathbf{h}}(\mathcal{S})$ with $g = \mathrm{clip}((|\mathcal{V}|-50)/50, 0, 1)$, and reads one head from it; repeat routes the short head output through the long head so that the two-branch pipeline exists but carries no extra span information.

Table II gives an unambiguous ordering, long < dual < short < repeat, the same on both fleets and in all four folds. Three readings follow.

History beyond the last 50 cycles is worth 5–6%. That is the difference between short and long, and it is the direct measurement of what masked pooling over the full valid span buys.

Fusing the spans is a net loss of 2–6%. dual is strictly worse than long on both fleets. The gate is not mis-calibrated—it is a deterministic function of the observable length—so the loss comes from mixing a dominated estimate into a good one. The short-span head is a useful regulariser during training and a harmful expert at test time.

The two-branch pipeline itself is expensive. repeat carries no additional span information yet is 18–39% worse, which is what one expects when a redundant branch doubles the path length through the network.

D. Where the residual error sits

The ensemble over-predicts by 1.6–2.4 cycles on average, and the bias is not uniform. It is close to zero for engines near failure and for engines with more than 100 cycles left, and peaks at +5 to +10 cycles when the true RUL is 60–90 cycles. That is the interval in which the model has seen enough history to become confident and not yet enough degradation to be corrected by the data.

This is the honest limitation of the method. The asymmetric penalty in (3) is meant to counter exactly this, and the NASA score—which prices over-prediction ten times more heavily—stays between 1.75 and 2.61, so the tail is controlled even though the mean is optimistic.

TABLE III
Ensemble metrics with 95% engine-clustered bootstrap intervals.

Fleet	Engines	RMSE	95% CI
FD001	100	9.636	[7.575, 11.678]
FD002	259	10.751	[9.195, 12.313]
FD003	100	10.684	[7.971, 13.212]
FD004	248	11.192	[9.253, 13.206]

E. Why FD003

FD003 is the only fleet where the method does not lead. It is also the fleet with the largest fold-to-fold spread (standard deviation 1.00 against 0.35–0.67 elsewhere) and the fleet where the two failure modes are drawn from a single operating condition. Our reading is that the long-span advantage measured in Table II is largest where histories are long and operating conditions vary (FD002, FD004), and that FD003 removes both sources of variation. We report the number as measured rather than tuning on it.

F. Statistical treatment

Confidence intervals in Table III resample engines, not engine-fold pairs, with 5000 bootstrap replicates. Resampling at the engine-fold level understates the interval by roughly a third because the same engine reappears across folds.

The intervals are wide—4.1 cycles on FD001—because they are honest about engine-level sampling. Differences

between methods smaller than about one cycle on a 100-engine fleet are not resolvable at this sample size.

G. Uncertainty quantification

A point estimate is not enough for maintenance scheduling, so we also measure whether the ensemble knows when it is wrong. Two interval constructions are compared: raw ensemble sigma, $\hat{y} \pm z\hat{\sigma}$ with $\hat{\sigma}$ the sample standard deviation over the fifteen models, and split conformal calibration (CQR) whose half-width is the $(1-\alpha)$ quantile of absolute residuals on the held-out folds.

Raw ensemble intervals under-cover badly: a nominal 90% interval covers 61–74% of engines, and a nominal 95% interval covers 69–80%. Fifteen models that share an architecture, a dataset, and a training protocol make highly correlated errors, so their spread systematically understates the true error. CQR restores coverage to 0.82–0.88 at the 90% level and 0.90–0.94 at 95%, but the intervals are 25–31 cycles wide at 90% and 34–45 at 95%, against a target range of 0–125. The cause is the heavy tail already visible in the bias analysis: the largest single error exceeds 60 cycles.

The honest reading is that this model is strong on point accuracy and weak on uncertainty. A 25-cycle interval does not support a maintenance decision. We report the numbers rather than a calibration trick that hides them.

H. Where the span optimum sits

The choice $L = 100$ is not arbitrary and not a monotone preference for more history. Two ablations bracket it. Against a 50-step pool the 100-step model wins 5.1–6.4% on FD002 and FD004 (Table II). Extending the input window to 200 steps instead loses 1.88% on FD003 validation (folds: −0.10, +0.47, +4.09, −1.52, +6.92%), and enlarging the encoder to $d = 96$ with three layers loses 4.8% on every one of the five FD001 folds.

Every training engine is longer than 100 cycles (mean 206 on FD001, 247 on FD003; 92–98% exceed 150), so the 200-step arm is not starved of data. The non-monotonicity is therefore about optimisation, not information: a longer sequence dilutes self-attention over positions whose marginal value is small, since degradation concentrates at the end of life. We report the bracket rather than a single point, because it constrains the claim: the method uses about a hundred steps, not "all available history".

I. Instance normalisation as an ensemble-diversity device

One further ablation is worth reporting because its sign is determined by a property of the fleet rather than by tuning. Adding a fourth input channel, the window differenced from its own engine baseline and divided by that engine's standard deviation, leaves single-model accuracy essentially unchanged (FD001 −0.50%, FD003 +0.15%, FD002 +1.46% on official test) but decorrelates the errors. Five such members added to the fifteen-model ensemble lower FD001 from 9.64 to 9.47 and FD003 from 10.68 to 10.60.

The same channel is catastrophic on FD004 (validation RMSE 14.85 against 9.5 for the baseline) and does not help FD002. The reason is mechanical: on fleets with six operating conditions the per-engine variance is dominated by condition switching, which is an order of magnitude larger than the degradation drift, so dividing by it compresses precisely the signal the model needs. On single-condition fleets the per-engine variance is mostly inter-unit nuisance, and removing it helps. The decision rule is therefore conditioned on the number of operating conditions, which is known before any test data is seen.

V. Discussion

The result we did not expect is that supervising two spans helps while fusing them hurts. A natural explanation is that the short head acts as a curriculum: it forces the encoder to keep recent-step information legible, which the long-head loss alone would allow to decay. At inference the long head already contains that information, and adding the short head separate estimate only injects its larger variance.

If that reading is right, the design rule generalises: use auxiliary spans as supervision, never as ensemble members. We have not tested it beyond C-MAPSS, and the two fleets where the effect is largest are also the ones with the most operating-condition variation.

VI. Conclusion

A shared encoder, a distance-from-end positional table, and masked mean pooling over the full valid history produce official-test RMSE of 9.64, 10.75, 10.68, and 11.19 cycles on C-MAPSS FD001–FD004 with fifteen-model ensembles, 7.6–20.7% below a 2026 state-of-the-art baseline on three fleets and 1.9% above it on the fourth. The ablation that produced this configuration also produced two negative results: fusing short and long spans costs 2–6%, and a redundant second branch costs 18–39%. The remaining error is concentrated as a positive bias of 5–10 cycles for engines at 60–90 cycles of remaining life, which is the interval where maintenance decisions are hardest.

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